10-02 第2篇

阅读理解（说明文） Why Do Leaves Turn Red and Yellow in Autumn?

① Every October, streets, parks and mountains in many parts of China slowly change colour. Green leaves turn yellow, orange, and sometimes a bright, fiery red. Tourists travel far just to take photos of red maple hills. But why does this happen? It is not magic — it is a neat piece of plant science.

② During spring and summer, leaves are **green** because they are full of a substance called *chlorophyll*. Chlorophyll helps a tree catch sunlight and turn it, together with water and air, into food. As long as a tree can make enough food, it keeps its leaves green and healthy. In those months, chlorophyll is so strong that it covers up every other colour inside the leaf.

③ But a leaf is not only green. Inside every leaf there are also yellow and orange colours, called *carotenoids*. They are there all year long — you just cannot see them because the green is too powerful. When autumn arrives, the weather grows colder and the days grow shorter. The tree senses these changes and begins to prepare for winter. Slowly, it stops making new chlorophyll, and the old chlorophyll breaks down. As the green fades, the hidden yellow and orange step forward at last. That is why ginkgo trees glow gold and birch forests turn pale yellow in October.

④ Red is a more interesting story. Red is **not** a hidden colour waiting inside the leaf. Instead, on cool, sunny autumn days, some trees — especially maples — actively make a new substance called *anthocyanin*, which is red or purple. Scientists are still studying why certain trees spend energy making red when they are about to drop their leaves. One popular idea is that the red colour protects the dying leaf from strong sunlight, so that the tree can pull back the last useful nutrients safely into its branches before the leaf falls.

⑤ Weather matters a lot. The brightest autumn colours appear after a stretch of warm, sunny days followed by cool, dry nights. If autumn is cloudy, or too warm, the red will be weaker. If an early frost comes, the leaves may turn brown and fall before they can show their best colour. That is why the "red-leaf season" is a little different every year, and why travel websites update "best viewing dates" again and again.

⑥ So next time you walk under a yellow ginkgo or pick up a scarlet maple leaf, remember: you are looking at a plant quietly saving energy for the long winter. The beauty of autumn is, in the end, the beauty of a living thing getting ready to rest.

注释： chlorophyll 叶绿素；carotenoid 类胡萝卜素；anthocyanin 花青素；frost 霜；scarlet 猩红色的

根据文章内容，选择最佳答案。

1. Why are leaves usually green in spring and summer?

A. Because trees paint their leaves green every year.
B. Because leaves contain a lot of chlorophyll, which is so strong that it covers all other colours.
C. Because the sunlight in summer is green.
D. Because yellow and orange colours have not yet formed inside the leaf.

2. According to Paragraph 3, why do leaves turn yellow in autumn?

A. New yellow chemicals suddenly appear because of the cold wind.
B. Yellow and orange colours are always in the leaf, but only show up after the green chlorophyll fades.
C. The tree catches yellow dust from the air in autumn.
D. Yellow leaves have been painted by rain and frost.

3. What is the main difference between yellow and red autumn colours?

A. Yellow colour is always inside the leaf; red is actively made by the tree in autumn with new anthocyanin.
B. Yellow is made in autumn; red has been hidden in the leaf all year.
C. Both colours appear only after the first snow.
D. Only cold mountain trees can show either colour.

4. Why might a tree make red anthocyanin in autumn, according to scientists?

A. To attract more birds for the winter.
B. To protect the dying leaf from strong sunlight, so the tree can safely take back useful nutrients from the leaf.
C. To become taller before winter.
D. To make tourists happy.

5. Which kind of autumn weather usually brings the brightest colours?

A. Many rainy, warm days in a row.
B. A sudden heavy snow in early October.
C. Warm, sunny days followed by cool, dry nights.
D. Long cloudy weeks with no sunshine at all.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段 "leaves are green because they are full of a substance called chlorophyll ... chlorophyll is so strong that it covers up every other colour inside the leaf."
【解析】绿色源自叶绿素，它掩盖了其他颜色。A "树给自己上色"、C "阳光是绿的"、D "黄橙色未形成"均不符原文（第③段明确说黄橙色一直都在）。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第③段 "yellow and orange colours ... are there all year long ... As the green fades, the hidden yellow and orange step forward at last."
【解析】黄橙色本就存在，绿色退去后才显露。A "突然出现"、C "空气中灰尘"、D "被雨霜染"均无依据。故选 B。

3.
【答案】A
【题型】推理判断题
【定位】第③段（黄色一直存在）+ 第④段 "Red is not a hidden colour ... trees ... actively make a new substance called anthocyanin."
【解析】黄色本存、秋显；红色需秋季主动合成。B 颠倒关系、C "初雪后"无依据、D "仅高山树"以偏概全。故选 A。

4.
【答案】B
【题型】细节理解题
【定位】第④段 "the red colour protects the dying leaf from strong sunlight, so that the tree can pull back the last useful nutrients ... before the leaf falls."
【解析】科学家推测红色保护叶片让树回收养分。A "吸引鸟"、C "长高"、D "取悦游客"均无依据。故选 B。

5.
【答案】C
【题型】细节理解题
【定位】第⑤段 "The brightest autumn colours appear after a stretch of warm, sunny days followed by cool, dry nights."
【解析】原文明确列出理想天气组合。A "连日阴雨"、B "早雪"、D "连周阴天"均不符。故选 C。